

Data warehouse implementation: Extraction, Transformation & Loading (ETL)

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Data Warehouse Design Recap: The Star Schema
Why Do We Still Use the Star Schema?

Core Idea: The Star Schema is a fundamental data warehouse design that supports efficient analytical queries while remaining simple and intuitive.

Key Components

Fact Table

- Stores business transactions and measures
- Contains mostly numeric values (e.g., sales amount, quantity)
- Links to dimension tables via foreign keys

Dimension Tables

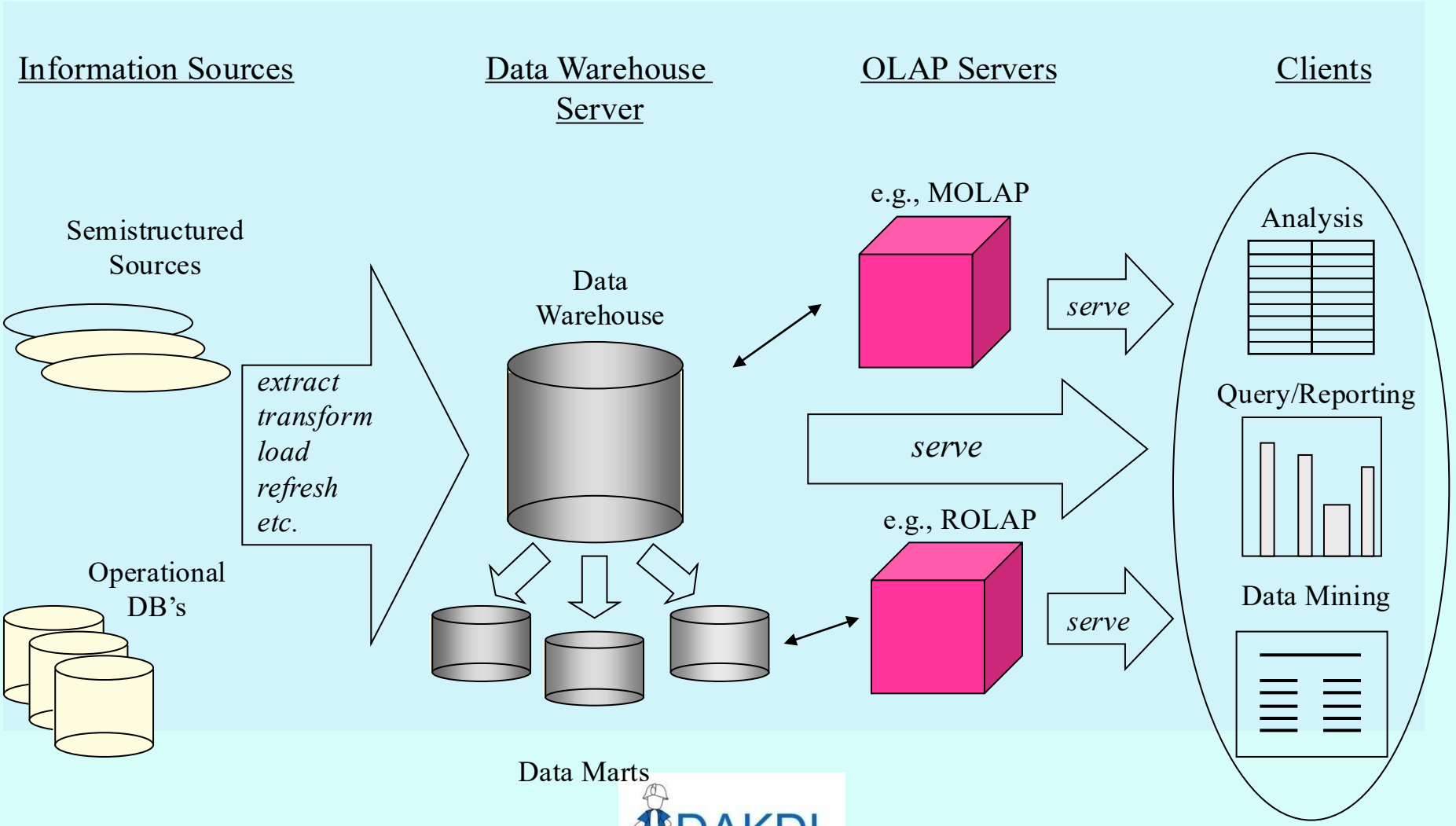
- Store descriptive attributes that provide context
- Answer analytical questions such as who, what, where, and when
- Examples: Time, Product, Customer, Location

Why the Star Schema Is Still Important

- Fast query performance (optimized for OLAP workloads)
- Simple and intuitive structure (easy to understand and maintain)
- Clear relationships between facts and dimensions
- Well supported by BI tools and dashboards

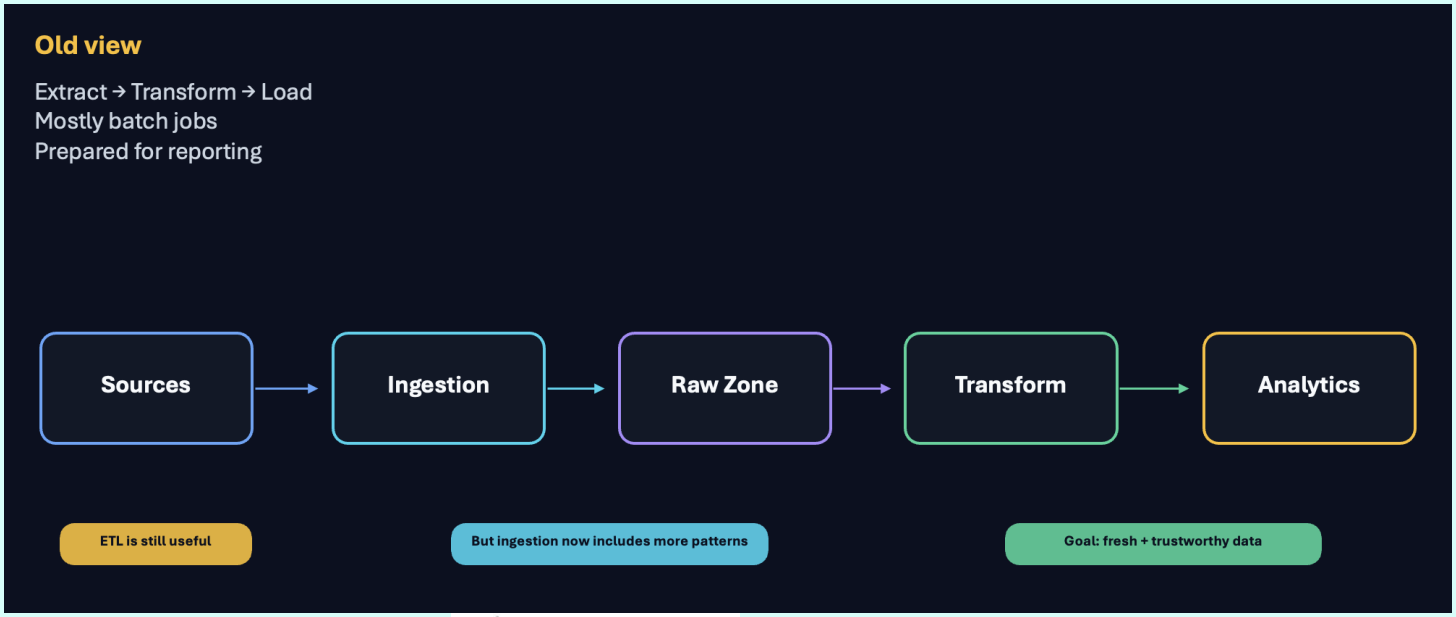


The Basic Elements of DW



ETL (Extraction, Transformation & Loading)

- **ETL** is a process that
  - **reads** data from one or more sources,
  - **transforms** the data to make it compatible with the destination system,
  - **loads** it into the target data warehouse.



ETL Main Tasks

- **Data Extraction**: Extract data from various source systems
- **Data Cleansing**: Clean data for clarification, meaningfulness, and accuracy
- **Data Transformation**: Transform source data into the format and structure for analytics
- **Data Integration**: Integrate & map data from all selected sources to the defined analytical data
- **Metadata Management**: Update metadata used for ETL mappings & rules
- **Measure deviation & Summarization**: Calculate measures based on defined formula
- **Incremental & Continuous Data Loading**: Iterative approach
- Schedule Run
- Data Quality
- Etc.



ETL process

- Consumes 70-80% of project time
- Heterogeneous Source Systems
- Little or no control over source systems
- Source systems scattered
- Different currencies, measurement units
- Ensuring data quality
- Typical functions
  - Define source, query (run SQL),
  - Define transformation, define target,
  - Verify transformation,
  - Schedule run, audit report

Data Sources: Lots of Dirty Data

Sales Information

Sales_log

time_tag	pid	sid	cid	price	qty
T01	Q33	A01	9901	10	2
T01	Q33	A01	9901	10	3
T01	K21	B01	9902	30	2
T01	Q33	A03	9903	10	1
T01	K21	A02	9903	5	2
T01	K21	A02	9903	5	3
T01	K21	A02	9903	5	4
T01	K21	A02	9903	5	1
T02	Q33	A01	9902	20	5
T02	K21	A03	9901	5	1
T02	K21	B02	9904	20	1
T04	K21	A02	9901	3	1
T05	P67	A01	9905	4	1
T05	K21	A02	9905	5	1

Cost

time_tag	pid	sid	unit_cost
T01	Q33	A01	5
T01	K21	B01	20
T01	Q33	A03	5
T01	K21	A02	2
T02	Q33	A01	10
T02	K21	A03	3
T02	K21	B02	15
T04	K21	A02	1
T05	P67	A01	3
T05	K21	A02	3

Store Information

Store_Desc

id	name	zip	region	state	phone
A01	Farwest	78700	Travis	TX	512-345-6678
A02	Anderson	78700	Travis	TX	512-342-3356
A03	Koenig	79220	Austin	TX	512-399-1245
B01	South	10020	Soho	NY	212-245-4563
B02	Central	10032	Midtown	NY	212-362-2278

Manager

sid	name
A01	Brown
A02	Molly
A03	Jones
B01	Jane
B02	Marvin

Customer_Desc

cid	name	age	sex	education	income	address
9901	David	30	1	1	80,000	34 Greystone, Austin TX 78730
9902	Nathan	55	1	1	100,000	23 Wood #21, Houston TX 71010
9903	Jane	55	2	2	95,000	12 Central, New York NY 10200
9904	Mary	23	2	3	60,000	9 King, Buffalo NY 11200
9905	Steve	25	1	3	55,000	11 Main, San Antonio TX 70340

Customer Information

Item_Code

pcode	name	brand
K21	Power_Clean	Cleaners
Q33	Coke_Classic	Cola
Q34	Coke_Diet	Cola
P67	Sprite	Coolers

School_Code

Code	Literal
1	Graduate
2	College
3	High School
4	Others

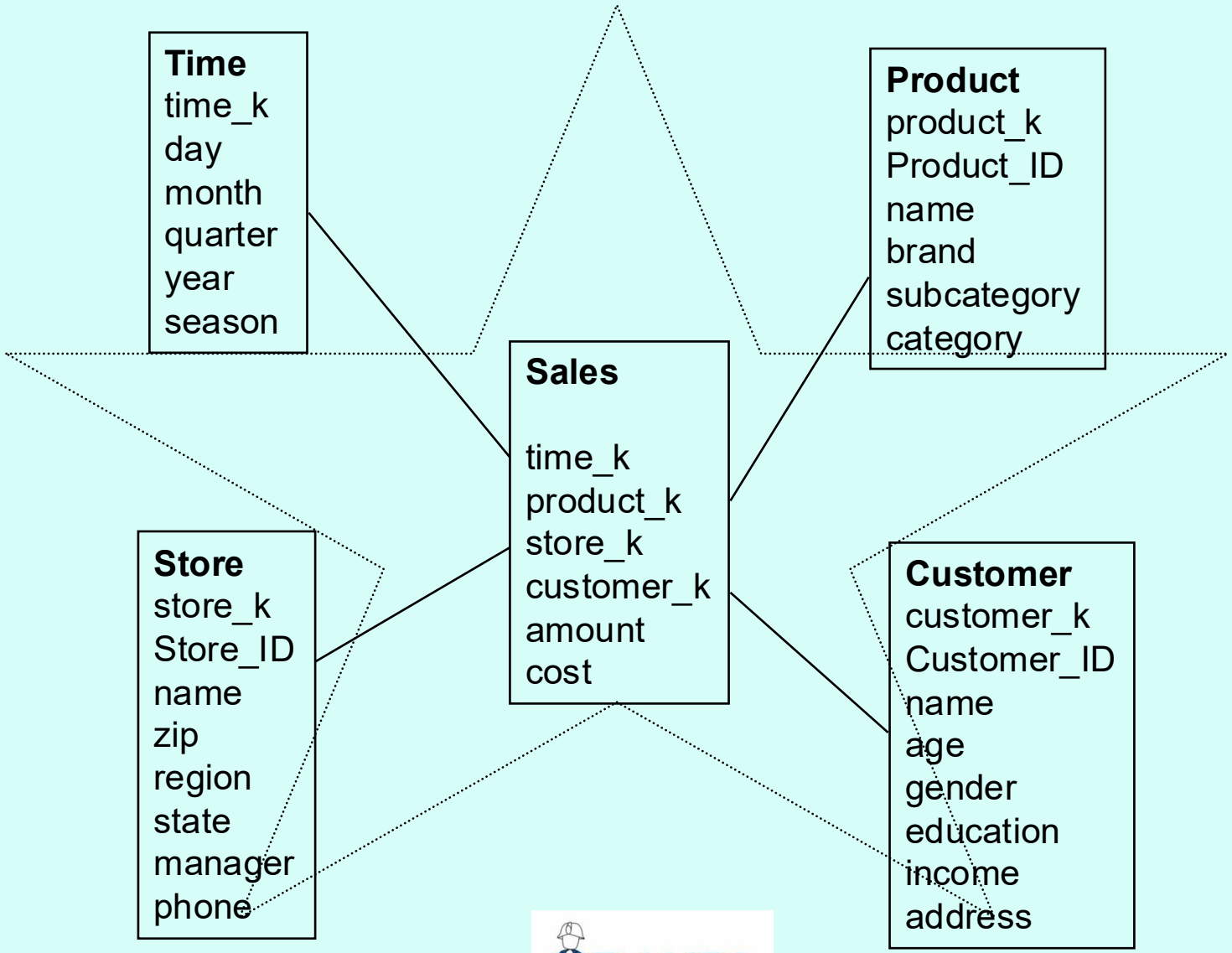
Gender_Code

Code	Literal
1	Male
2	Female

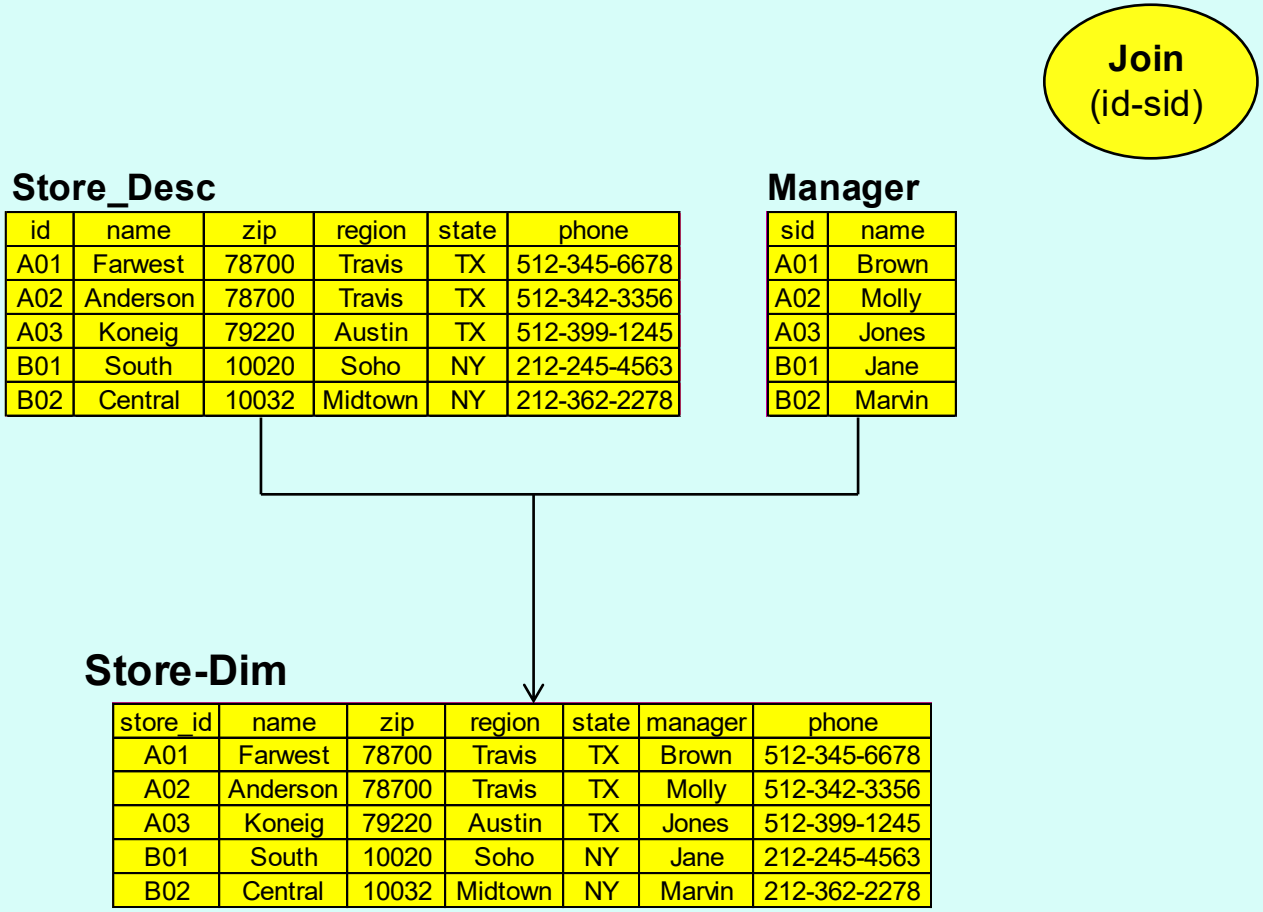
Code Information



Star Schema in a Data Warehouse



ETL: Building Store-Dim Table



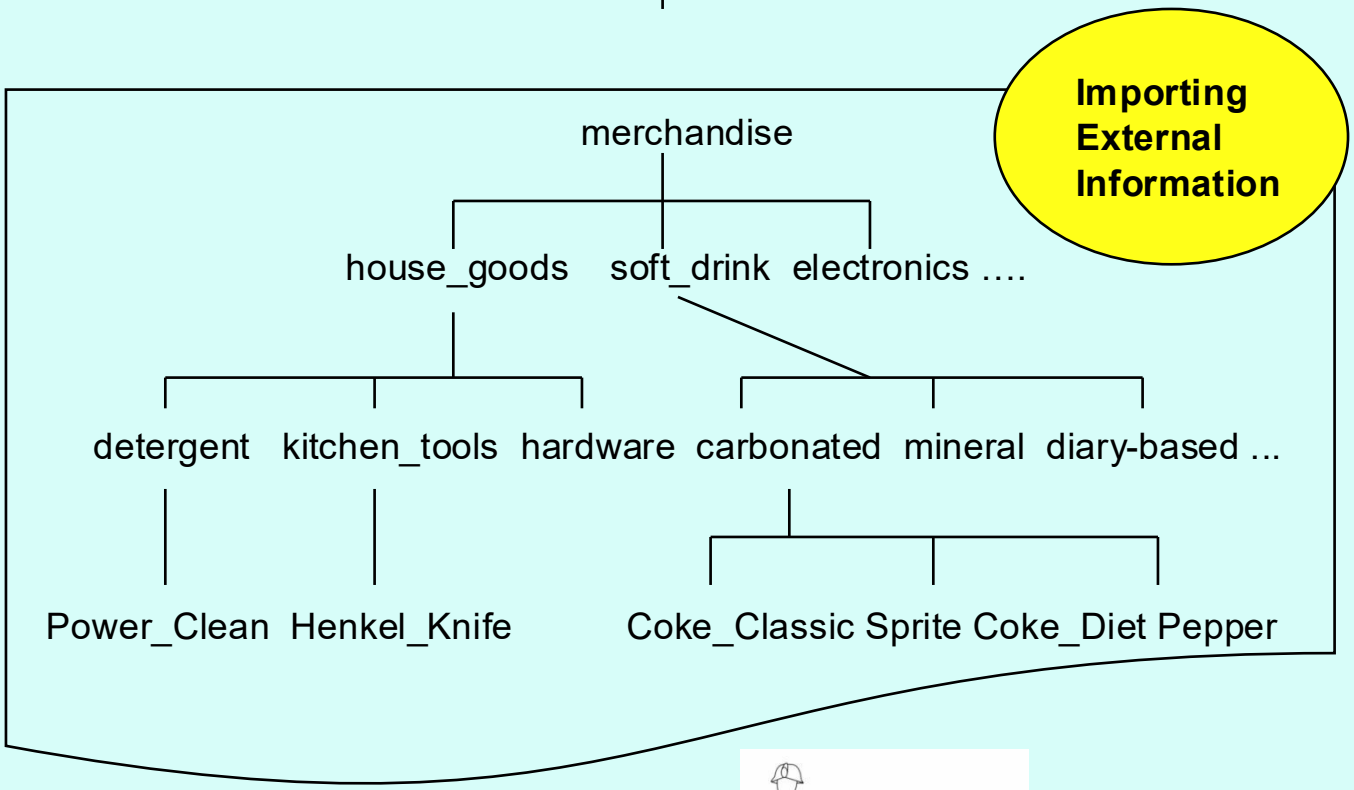
ETL: Building Product-Dim Table

Item_Code

pcode	name	brand
K21	Power_Clean	Cleaners
Q33	Coke_Classic	Cola
Q34	Coke_Diet	Cola
P67	Sprite	Coolers

Product

pcode	name	brand	subcategory	category
K21	Power_Clean	Cleaners	detergent	house_goods
Q33	Coke_Classic	Cola	carbonated	soft_drink
Q34	Coke_Diet	Cola	carbonated	soft_drink
P67	Sprite	Coolers	carbonated	soft_drink



ETL: Building Customer-Dim Table



Customer_Desc

cid	name	age	sex	education	income	address
9901	David	30	1	1	80,000	34 Greystone, Austin TX 78730
9902	Nathan	55	1	1	100,000	23 Wood #21, Houston TX 71010
9903	Jane	55	2	2	95,000	12 Central, New York NY 10200
9904	Mary	23	2	3	60,000	9 King, Buffalo NY 11200
9905	Steve	25	1	3	55,000	11 Main, San Antonio TX 70340

Education_Code

Code	Literal
1	Graduate
2	College
3	High School
4	Others

Gender_Code

Code	Literal
1	Male
2	Female

Customer-Dim

cid	name	age	gender	education	income	address
9901	David	30	Male	Graduate	80,000	34 Greystone, Austin TX 78730
9902	Nathan	55	Male	Graduate	100,000	23 Wood #21, Houston TX 71010
9903	Jane	55	Female	College	95,000	12 Central, New York NY 10200
9904	Mary	23	Female	High_School	60,000	9 King, Buffalo NY 11200
9905	Steve	25	Male	High_School	55,000	11 Main, San Antonio TX 70340

DAKDL
Data Analysis & Knowledge Discovery Lab

ETL: Building Sales-Fact Table (1/3)

Sales_log

time_tag	pid	sid	cid	price	qty
T01	Q33	A01	9901	10	2
T01	Q33	A01	9901	10	3
T01	K21	B01	9902	30	2
T01	Q33	A03	9903	10	1
T01	K21	A02	9903	5	2
T01	K21	A02	9903	5	3
T01	K21	A02	9903	5	4
T01	K21	A02	9903	5	1
T02	Q33	A01	9902	20	5
T02	K21	A03	9901	5	1
T02	K21	B02	9904	20	1
T04	K21	A02	9901	3	1
T05	P67	A01	9905	4	1
T05	K21	A02	9905	10	1

Cost

time_tag	pid	sid	unit_cost
T01	Q33	A01	5
T01	K21	B01	20
T01	Q33	A03	5
T01	K21	A02	2
T02	Q33	A01	10
T02	K21	A03	3
T02	K21	B02	15
T04	K21	A02	1
T05	P67	A01	3
T05	K21	A02	3

Join
(time_tag, pid, sid)

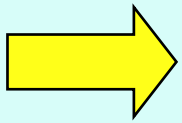
time_tag	pid	sid	cid	price	qty	unit_cost
T01	Q33	A01	9901	10	2	5
T01	Q33	A01	9901	10	3	5
T01	K21	B01	9902	30	2	20
T01	Q33	A03	9903	10	1	5
T01	K21	A02	9903	5	2	2
T01	K21	A02	9903	5	3	2
T01	K21	A02	9903	5	4	2
T01	K21	A02	9903	5	1	2
T02	Q33	A01	9902	20	5	10
T02	K21	A03	9901	5	1	3
T02	K21	B02	9904	20	1	15
T04	K21	A02	9901	3	1	1
T05	P67	A01	9905	4	1	3
T05	K21	A02	9905	10	1	3

DAKDL
Data Analysis & Knowledge Discovery Lab

ETL: Building Sales-Fact Table (2/3)

Computation
amount = price * qty
cost = unit_cost * qty

time_tag	pid	sid	cid	price	qty	unit_cost
T01	Q33	A01	9901	10	2	5
T01	Q33	A01	9901	10	3	5
T01	K21	B01	9902	30	2	20
T01	Q33	A03	9903	10	1	5
T01	K21	A02	9903	5	2	2
T01	K21	A02	9903	5	3	2
T01	K21	A02	9903	5	4	2
T01	K21	A02	9903	5	1	2
T02	Q33	A01	9902	20	5	10
T02	K21	A03	9901	5	1	3
T02	K21	B02	9904	20	1	15
T04	K21	A02	9901	3	1	1
T05	P67	A01	9905	4	1	3
T05	K21	A02	9905	10	1	3

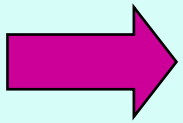


time_tag	pid	sid	cid	amount	cost
T01	Q33	A01	9901	20	10
T01	Q33	A01	9901	30	15
T01	K21	B01	9902	60	40
T01	Q33	A03	9903	10	5
T01	K21	A02	9903	10	4
T01	K21	A02	9903	15	6
T01	K21	A02	9903	20	8
T01	K21	A02	9903	5	2
T02	Q33	A01	9902	100	50
T02	K21	A03	9901	5	3
T02	K21	B02	9904	20	15
T04	K21	A02	9901	3	1
T05	P67	A01	9905	4	3
T05	K21	A02	9905	10	3

ETL: Building Sales-Fact Table (3/3)

Grouping (time_tag, pid, sid, cid) **and Aggregation** (amount, cost)

time_tag	pid	sid	cid	amount	cost
T01	Q33	A01	9901	20	10
T01	Q33	A01	9901	30	15
T01	K21	B01	9902	60	40
T01	Q33	A03	9903	10	5
T01	K21	A02	9903	10	4
T01	K21	A02	9903	15	6
T01	K21	A02	9903	20	8
T01	K21	A02	9903	5	2
T02	Q33	A01	9902	100	50
T02	K21	A03	9901	5	3
T02	K21	B02	9904	20	15
T04	K21	A02	9901	3	1
T05	P67	A01	9905	4	3
T05	K21	A02	9905	10	3



Sales-Fact

time_k	product_k	store_k	customer_k	amount	cost
T01	Q33	A01	9901	50	25
T01	K21	B01	9902	60	40
T01	Q33	A03	9903	10	5
T01	K21	A02	9903	50	20
T02	Q33	A01	9902	100	50
T02	K21	A03	9901	5	3
T02	K21	B02	9904	20	15
T04	K21	A02	9901	3	1
T05	P67	A01	9905	4	3
T05	K21	A02	9905	10	3

DAKDL Data Analysis & Knowledge Discovery Lab

Data Warehouse Dim & Fact Tables

Time

time_k	day	month	quarter	year	season
T01	1	Jan.	1Q	1999	New Years
T02	2	Jan.	1Q	1999	New Years
T03	3	Jan.	1Q	1999	Normal
T04	4	Jan.	1Q	1999	Normal
T05	5	Jan.	1Q	1999	Normal

Customer

customer_k	name	age	gender	education	income	address
9901	David	30	Male	Graduate	80,000	34 Greystone, Austin TX 78730
9902	Nathan	55	Male	Graduate	100,000	23 Wood #21, Houston TX 71010
9903	Jane	55	Female	College	95,000	12 Central, New York NY 10200
9904	Mary	23	Female	High_School	60,000	9 King, Buffalo NY 11200
9905	Steve	25	Male	High_School	55,000	11 Main, San Antonio TX 70340

Sales

time_k	product_k	store_k	customer_k	amount	cost
T01	Q33	A01	9901	50	25
T01	K21	B01	9902	60	40
T01	Q33	A03	9903	10	5
T02	K21	A02	9903	50	20
T02	Q33	A01	9902	100	50
T02	K21	A03	9901	5	3
T02	K21	B02	9904	20	15
T04	K21	A02	9901	3	1
T05	P67	A01	9905	4	3
T05	K21	A02	9905	10	3

Product

product_k	name	brand	subcategory	category
K21	Power_Clean	Cleaners	detergent	house_goods
Q33	Coke_Classic	Cola	carbonated	soft_drink
Q34	Coke_Diet	Cola	carbonated	soft_drink
P67	Sprite	Coolers	carbonated	soft_drink

Store

store_k	name	zip	region	state	manager	phone
A01	Farwest	78700	Travis	TX	Brown	512-345-6678
A02	Anderson	78700	Travis	TX	Molly	512-342-3356
A03	Koneig	79220	Austin	TX	Jones	512-399-1245
B01	South	10020	Soho	NY	Jane	212-245-4563
B02	Central	10032	Midtown	NY	Marvin	212-362-2278

SQL on Dim Table

Customer

customer_k	name	age	gender	education	income	address
9901	David	30	Male	Graduate	80,000	34 Greystone, Austin TX 78730
9902	Nathan	55	Male	Graduate	100,000	23 Wood #21, Houston TX 71010
9903	Jane	55	Female	College	95,000	12 Central, New York NY 10200
9904	Mary	23	Female	High_School	60,000	9 King, Buffalo NY 11200
9905	Steve	25	Male	High_School	55,000	11 Main, San Antonio TX 70340

- **Find customers who are more than 50 years old, and earn more than $70,000**

```
SELECT *
FROM Customer
WHERE age > 50 AND income > 70000;
```

Result

customer_k	name	age	gender	education	income	address
9902	Nathan	55	Male	Graduate	100,000	23 Wood #21, Houston TX 71010
9903	Jane	55	Female	College	95,000	12 Central, New York NY 10200

SQL on Dim & Fact Tables

- **Find the total amount of sales for each brand of soft drink?**

```
SELECT Product.brand, sum(Sales.amount)
FROM Product, Sales
WHERE Sales.product_k = Product.product_k
AND Product.category = 'soft_drink'
GROUP BY Product.brand;
```

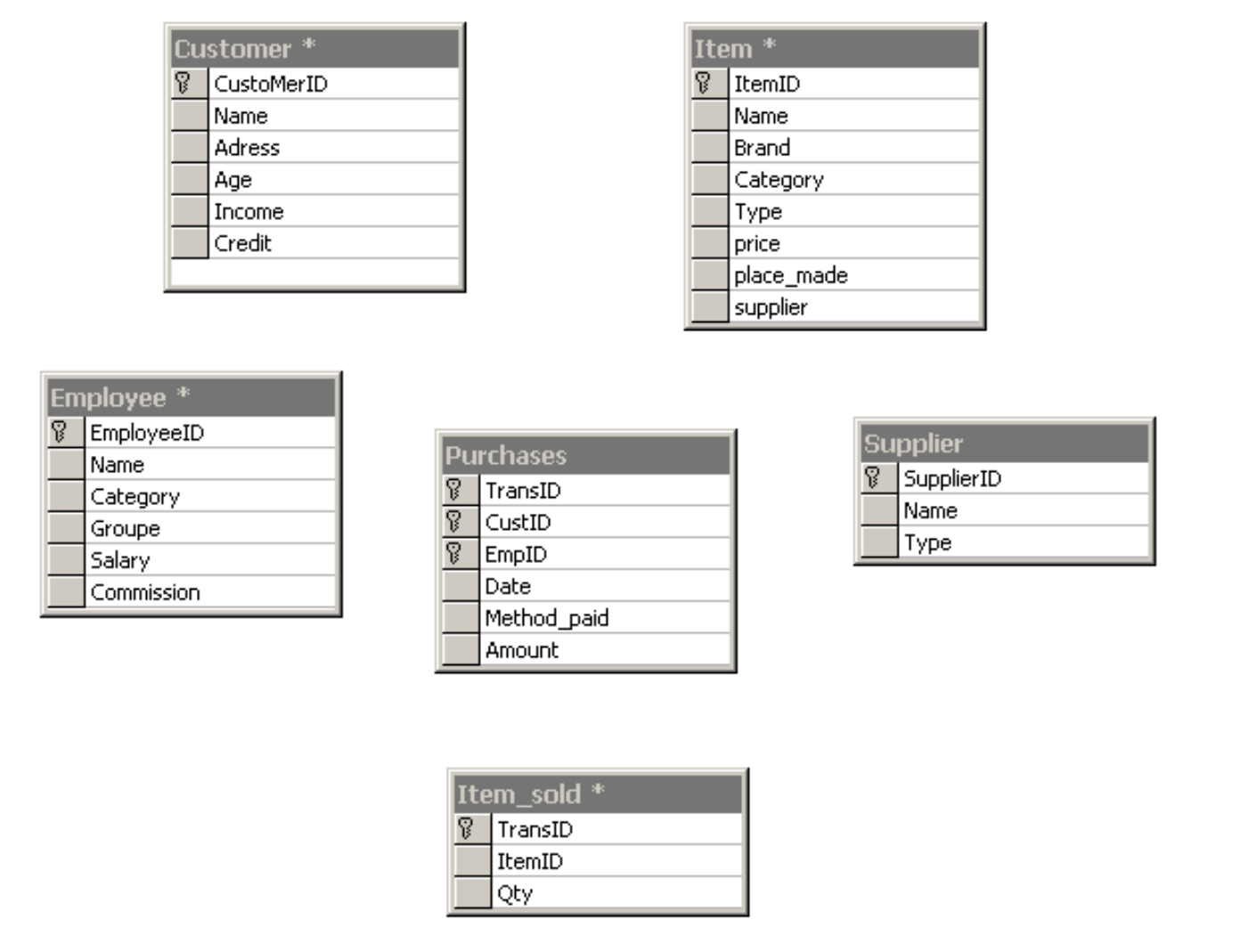
Result

brand	amount
Cola	160
Coolers	4

Some ETL Tools

- DataStage
- Informatica
- SAS ETL Server
- Information Builder's Data Migrator/ETL Manager tool suite
- Sunopsis
- Sqlser-Server BI
- Power-BI
- Rapid-Miner
- Etc.

AllElectronics-OLTP



Requirements

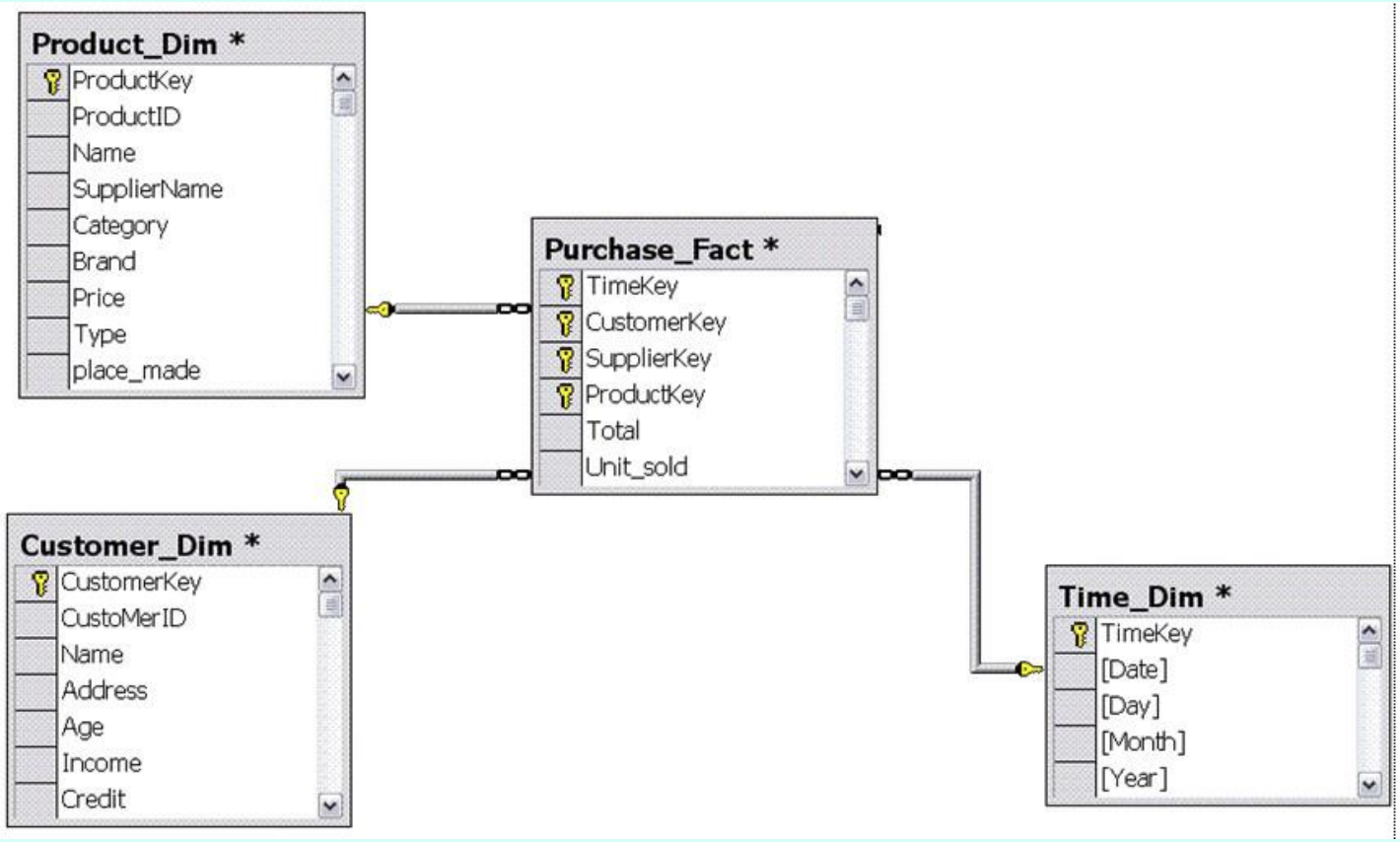
- What products were sold?
- Which customers bought what products?
- Characteristics of buyers for a particular product?
- When products were purchased? At what quantity?
- Total sales for each product category for a particular customer
- …

4-step design

Dimensional Design Process

- **Step1. Select the business process**
  - **"Sales Analysis" business process to analyze**
    - What products were sold? At which conditions?
    - Which customers bought what products?
    - Characteristics of buyers for a particular product?
    - When products were purchased? At what quantity?
    - Total sales for each product category for a particular customer
- **Step2. Declare the grain**
  - **Daily sale by product and customer**
- **Step3. Choose the dimensions**
  - **Date, Product and Customer**
- **Step4. Identify the facts**
  - **Total Sales, Quantity Sold**

AllElect Star-Schema



Star-schema dimension tables creation [MySQL]

- Step 1: Create dimension tables with **surrogate keys**

```
-- Customer Dimension
CREATE TABLE CustomerDim
(
CustomerKey INT AUTO_INCREMENT NOT NULL,
CustomerID INT NOT NULL,
CustomerName VARCHAR(50) NOT NULL,
CustomerAge INT NOT NULL,
CustomerIncome DOUBLE NOT NULL,
CustomerCredit VARCHAR(50) NOT NULL,
PRIMARY KEY (CustomerKey) );
```

```
-- Product Dimension
CREATE TABLE ProductDim
( ProductKey INT AUTO_INCREMENT NOT NULL,
ProductID INT NOT NULL,
ProductName VARCHAR(50) NOT NULL,
ProductCategory VARCHAR(50) NOT NULL,
SupplierName VARCHAR(50) NOT NULL,
ProductBrand VARCHAR(50) NOT NULL,
ProductPrice DECIMAL(10, 2) NOT NULL,
ProductType VARCHAR(50) NOT NULL,
ProductPlace_made VARCHAR(50) NOT NULL,
PRIMARY KEY (ProductKey) );
```

```
-- Date Dimension
CREATE TABLE DateDim
(
DateKey INT AUTO_INCREMENT NOT NULL,
Day INT NOT NULL,
Month VARCHAR(50) NOT NULL,
Year VARCHAR(50) NOT NULL,
TheDate DATETIME NOT NULL,
PRIMARY KEY (DateKey) );
```



Star-schema date dimension tables creation [MySQL]

```
CREATE TABLE DateDim (
    DateKey INT AUTO_INCREMENT NOT NULL,
    TheDate DATE NOT NULL,
    FullDate DATETIME NOT NULL,

    -- Day attributes
    Day INT NOT NULL,
    DayOfWeek INT NOT NULL,
    DayOfWeekName VARCHAR(20) NOT NULL,
    DayOfWeekShort VARCHAR(3) NOT NULL,
    DayOfYear INT NOT NULL,

    -- Week attributes
    WeekOfYear INT NOT NULL,
    WeekOfMonth INT NOT NULL,

    -- Month attributes
    Month INT NOT NULL,
    MonthName VARCHAR(20) NOT NULL,
    MonthShort VARCHAR(3) NOT NULL,

    -- Quarter attributes
    Quarter INT NOT NULL,
    QuarterName VARCHAR(10) NOT NULL,

    -- Year attributes
    Year INT NOT NULL,

    -- Business day flags
    IsWeekday BOOLEAN NOT NULL,
    IsWeekend BOOLEAN NOT NULL,
    IsHoliday BOOLEAN DEFAULT FALSE,

    -- Fiscal period (assuming fiscal year starts in January,
adjust as needed)
    FiscalYear INT NOT NULL,
    FiscalQuarter INT NOT NULL,
    FiscalMonth INT NOT NULL,

    -- Relative date helpers
    IsCurrentDay BOOLEAN DEFAULT FALSE,
    IsCurrentMonth BOOLEAN DEFAULT FALSE,
    IsCurrentYear BOOLEAN DEFAULT FALSE,

    -- Formatted dates
    DateFormatMMDDYYYY VARCHAR(10) NOT NULL,
    DateFormatDDMMYYYY VARCHAR(10) NOT NULL,
    DateFormatYYYYMMDD VARCHAR(10) NOT NULL,

    PRIMARY KEY (DateKey),
    UNIQUE KEY UK_TheDate (TheDate)
);
```



Star-schema dimension tables creation [SQL-Server]

- Step 1: Create dimension tables with surrogate keys

```
-- Customer Dimension
CREATE TABLE CustomerDim
(
CustomerKey int IDENTITY (1,1) NOT NULL,
CustoMerID int NOT NULL,
CustomerName nvarchar(50) NOT NULL,
CustomerAge int NOT NULL,
CustomerIncome float NOT NULL,
CustomerCredit nvarchar(50) NOT NULL
);
-- Product Dimension
CREATE TABLE ProductDim
(
ProductKey int IDENTITY (1,1) NOT NULL,
ProductID int NOT NULL,
ProductName nvarchar(50) NOT NULL,
ProductCategory nvarchar(50) NOT NULL,
SupplierName nvarchar(50) NOT NULL,
ProductBrand nvarchar(50) NOT NULL,
ProductPrice money NOT NULL,
ProductType nvarchar(50) NOT NULL,
ProductPlace_made nvarchar(50) NOT NULL
);
```

```
-- Date Dimension
CREATE TABLE DateDim
(
DateKey int IDENTITY (1,1) NOT NULL,
Day int NOT NULL,
Month nvarchar(50) NOT NULL,
Year nvarchar(50) NOT NULL,
TheDate datetime NOT NULL
);
```

Star-schema fact table creation [MySQL]

Step 2: Create fact tables

```
CREATE TABLE ElectFact (
   ProductKey INT NOT NULL,
   CustomerKey INT NOT NULL,
   TimeKey INT NOT NULL,
   unit_sold INT NOT NULL,
   total DECIMAL(10, 2) NOT NULL,

   PRIMARY KEY (ProductKey, CustomerKey, TimeKey),

   CONSTRAINT FK_ElectFact_Product
      FOREIGN KEY (ProductKey) REFERENCES ProductDim(ProductKey),

   CONSTRAINT FK_ElectFact_Customer
      FOREIGN KEY (CustomerKey) REFERENCES CustomerDim(CustomerKey),

   CONSTRAINT FK_ElectFact_Date
      FOREIGN KEY (TimeKey) REFERENCES DateDim(DateKey)
);
-- Optional: Add indexes for better query performance
CREATE INDEX IDX_ElectFact_Product ON ElectFact(ProductKey);
CREATE INDEX IDX_ElectFact_Customer ON ElectFact(CustomerKey);
CREATE INDEX IDX_ElectFact_Date ON ElectFact(TimeKey);
```

Star-schema fact table creation [SQL-Server]

Step 2: Create fact tables

```
CREATE TABLE ElectFact
(
ProductKey int NOT NULL,
CustomerKey int NOT NULL,
TimeKey int NOT NULL,
unit_sold int NOT NULL,
total money NOT NULL
)

ALTER TABLE ElectFact WITH NOCHECK ADD
CONSTRAINT PK_Sales_fact
PRIMARY KEY NONCLUSTERED(ProductKey,CustomerKey,TimeKey)
```



Star-schema populate customer dimension [MySQL]

```
-- Populate CustomerDim from OLTP database AllElectronics.Customers

-- Populate Customer Dimension
INSERT INTO CustomerDim (CustomerID, CustomerName, CustomerAge,
CustomerIncome, CustomerCredit)
SELECT
    c.CustomerID,
    c.Name,
    c.Age,
    c.Income,
    c.Credit
FROM AllElectronics.Customers c;
```

```
CREATE TABLE CustomerDim
(
CustomerKey INT AUTO_INCREMENT NOT NULL,
CustomerID INT NOT NULL,
CustomerName VARCHAR(50) NOT NULL,
CustomerAge INT NOT NULL,
CustomerIncome DOUBLE NOT NULL,
CustomerCredit VARCHAR(50) NOT NULL,
PRIMARY KEY (CustomerKey) );
```

Star-schema populate product dimension [MySQL]

```
-- Populate CustomerDim from OLTP database AllElectronics.Customers

-- Populate ProductDim from AllElectronics.Items and Suppliers tables
-- Populate Product Dimension

INSERT INTO ProductDim (ProductID, ProductName, ProductCategory, SupplierName,
                        ProductBrand, ProductPrice, ProductType, ProductPlace_made)
SELECT
   i.ItemID,
   i.ItemName,
   i.Category,
   s.SupplierName,
   i.Brand,
   i.Price,
   i.Type,
   i.PlaceMade
FROM AllElectronics.Items i
INNER JOIN AllElectronics.Suppliers s
   ON i.SupplierID = s.SupplierID;
```

```
-- Product Dimension
CREATE TABLE ProductDim
( ProductKey INT AUTO_INCREMENT NOT NULL,
ProductID INT NOT NULL,
ProductName VARCHAR(50) NOT NULL,
ProductCategory VARCHAR(50) NOT NULL,
SupplierName VARCHAR(50) NOT NULL,
ProductBrand VARCHAR(50) NOT NULL,
ProductPrice DECIMAL(10, 2) NOT NULL,
ProductType VARCHAR(50) NOT NULL,
ProductPlace_made VARCHAR(50) NOT NULL,
PRIMARY KEY (ProductKey) );
```

Star-schema populate date dimension [MySQL]

```
-- Populate DateDim from AllElectronics.Purchases
INSERT INTO DateDim (TheDate, FullDate, Day, DayOfWeek, DayOfWeekName, DayOfWeekShort,
                     DayOfYear, WeekOfYear, WeekOfMonth, Month, MonthName, MonthShort,
                     Quarter, QuarterName, Year, IsWeekday, IsWeekend, IsHoliday,
                     FiscalYear, FiscalQuarter, FiscalMonth, IsCurrentDay, IsCurrentMonth,
                     IsCurrentYear, DateFormatMMDDYYYY, DateFormatDDMMYYYY, DateFormatYYYYMMDD)
SELECT DISTINCT
   DATE(p.Date) AS TheDate,
   p.Date AS FullDate,
   DAY(p.Date) AS Day,
   DAYOFWEEK(p.Date) AS DayOfWeek,
   DAYNAME(p.Date) AS DayOfWeekName,
   LEFT(DAYNAME(p.Date), 3) AS DayOfWeekShort,
   DAYOFYEAR(p.Date) AS DayOfYear,
   WEEK(p.Date, 3) AS WeekOfYear,
   CEILING(DAY(p.Date) / 7) AS WeekOfMonth,
   MONTH(p.Date) AS Month,
   MONTHNAME(p.Date) AS MonthName,
   LEFT(MONTHNAME(p.Date), 3) AS MonthShort,
   QUARTER(p.Date) AS Quarter,
   CONCAT('Q', QUARTER(p.Date)) AS QuarterName,
   YEAR(p.Date) AS Year,
   DAYOFWEEK(p.Date) NOT IN (1, 7) AS IsWeekday,
   DAYOFWEEK(p.Date) IN (1, 7) AS IsWeekend,
   FALSE AS IsHoliday,
   YEAR(p.Date) AS FiscalYear,
   QUARTER(p.Date) AS FiscalQuarter,
   MONTH(p.Date) AS FiscalMonth,
   DATE(p.Date) = CURDATE() AS IsCurrentDay,
```



Star-schema populate fact table [MySQL]

```
INSERT INTO ElectFact (ProductKey, CustomerKey, TimeKey, unit_sold, total)
SELECT
   pd.ProductKey,
   cd.CustomerKey,
   dd.DateKey,
   SUM(s.Qty) AS unit_sold,
   SUM(i.Price * s.Qty) AS total
FROM AllElectronics.Item_sold s
INNER JOIN AllElectronics.Purchases p ON s.TransID = p.TransID
INNER JOIN AllElectronics.Items i ON s.ItemID = i.ItemID
INNER JOIN ProductDim pd ON pd.ProductID = i.ItemID
INNER JOIN CustomerDim cd ON cd.CustomerID = p.CustID
INNER JOIN DateDim dd ON dd.TheDate = DATE(p.Date)
GROUP BY
   pd.ProductKey,
   cd.CustomerKey,
   dd.DateKey;
```

Star-schema populate fact table [SQL-Server]

```
INSERT INTO dbo.ElectFact (ProductKey, CustomerKey, TimeKey, unit_sold, total)
SELECT
    pd.ProductKey,
    cd.CustomerKey,
    dd.DateKey,
    SUM(s.Qty) AS unit_sold,
    SUM(i.Price * s.Qty) AS total
FROM AllElectronics.dbo.Item_sold s
JOIN AllElectronics.dbo.Purchases p ON s.TransID = p.TransID
JOIN AllElectronics.dbo.Items i ON s.ItemID = i.ItemID

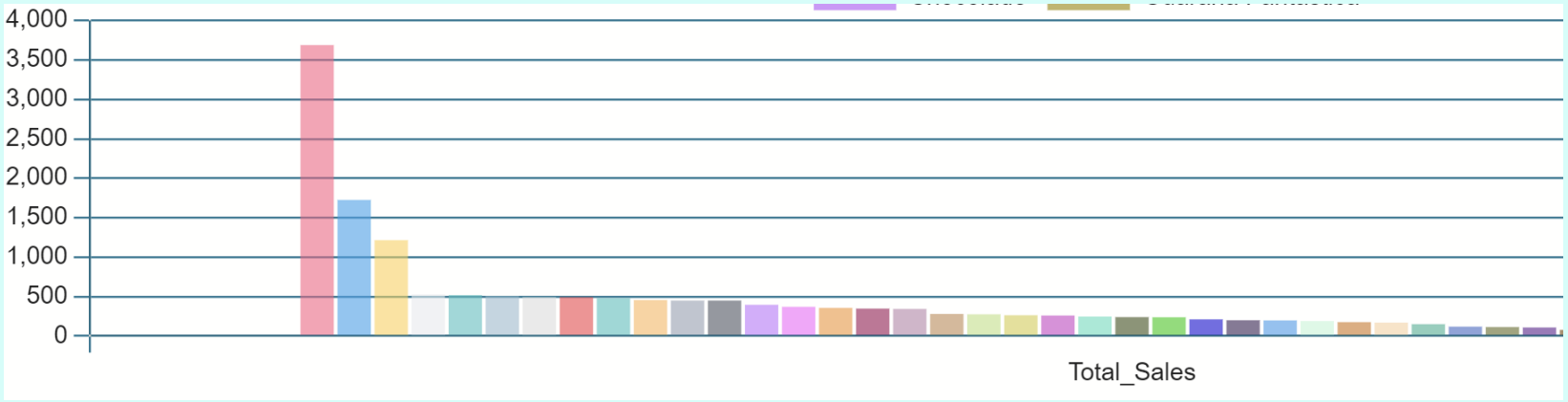
JOIN dbo.ProductDim pd ON pd.ProductID = i.ItemID
JOIN dbo.CustomerDim cd ON cd.CustoMerID = p.CustID
JOIN dbo.DateDim dd ON dd.TheDate = CAST(p.Date AS datetime)
GROUP BY
    pd.ProductKey,
    cd.CustomerKey,
    dd.DateKey;
```

Total sales for each product

```
SELECT
    pd.ProductName,
    SUM(f.total) AS Total_Sales
FROM dbo.ElectFact f
JOIN dbo.ProductDim pd ON f.ProductKey = pd.ProductKey
GROUP BY pd.ProductName
ORDER BY Total_Sales DESC;
```

Total sales for each product (in DESC order)

```
SELECT
    pd.ProductName,
    SUM(f.total) AS Total_Sales
FROM dbo.ElectFact f
JOIN dbo.ProductDim pd ON f.ProductKey = pd.ProductKey
GROUP BY pd.ProductName
ORDER BY Total_Sales DESC;
```



Total sales of each product category specifically for products supplied by "Exotic Liquids"

```
SELECT
    pd.ProductCategory,
    SUM(f.total) AS Total_Sales
FROM dbo.ElectFact f
JOIN dbo.ProductDim pd ON f.ProductKey = pd.ProductKey
WHERE pd.SupplierName = 'Exotic Liquids'
GROUP BY pd.ProductCategory
ORDER BY Total_Sales DESC;
```

